

Lecture 4

Frequentist versus Bayesian

Arnab Hazra

(Edited from the lecture notes by Brian J. Reich, NC State)



A motivating example

- ▶ Suppose, you are interested in balloon shooting in a village fair.
- ▶ Say, you shoot $n = 100$ times and hit $Y = 60$ balloons.
- ▶ A reasonable model for the number of hits would be $Y \sim \text{Binomial}(n, \theta)$.
- ▶ The parameter $\theta \in [0, 1]$ is the true probability of a successful shooting.
- ▶ What's your best guess about θ ?
- ▶ What's the probability that θ is greater than a half?

Frequentist approach

- ▶ A **frequentist procedure** quantifies uncertainty in terms of *repeating the process that generated the data many times*.
- ▶ The parameters θ are fixed and unknown.
- ▶ The sample (data) Y is random.
- ▶ A frequentist would **never** say $\text{Prob}(\theta > 0.5) = 0.60$ because θ is not a random variable.
- ▶ All probability statements should be made about randomness in the data.

Frequentist approach

- ▶ A **frequentist** procedure quantifies uncertainty in terms of *repeating the process that generated the data many times*.
- ▶ For an illustration see <https://www.rossmanchance.com/applets/2021/confsim/ConfSim.html>
- ▶ A **statistic** $\hat{\theta}$ is a summary of the sample.
- ▶ For example, the sample proportion $\hat{\theta} = Y/n$ is a statistic, and it is an **estimator** of the true proportion θ .
- ▶ The distribution of $\hat{\theta}$ that arises from repeating the process that generated the data many times is its **sampling distribution**.
- ▶ A frequentist would **never** say “the distribution of θ is Beta(6,4)”.

Frequentist approach

- ▶ A **frequentist** procedure quantifies uncertainty in terms of *repeating the process that generated the data many times*.
- ▶ A **95% confidence interval** (l, u) is

If we repeat the experiment 100,000 times, then, on an average, the true unknown θ falls within (l, u) for 95,000 of the times.

- ▶ A frequentist would **never** say “the probability that the true proportion is in the interval $(0.4, 0.8)$ is 0.95”.

Frequentist approach

- ▶ A **frequentist** procedure quantifies uncertainty in terms of *repeating the process that generated the data many times*.
- ▶ A common approach for testing a hypothesis is to reject the null if a test statistic exceeds a threshold.
- ▶ For example, we might reject $\mathcal{H}_0 : \theta \leq 0.5$ in favor of the alternative $\mathcal{H}_1 : \theta > 0.5$ if $\hat{\theta} = Y/n > T$.

- ▶ A **p-value** is

Under the null hypothesis, the probability that the test statistic $\hat{\theta} = Y/n$ exceeds the observed value, i.e., 0.6.

- ▶ A frequentist would **never** say “the probability that the null hypothesis is true is 0.03”.

Frequentist approach

There is currently an intense discussion of the merits of the p-value in the scientific community:

- ▶ <https://www.nature.com/articles/506150a>
- ▶ <https://fivethirtyeight.com/features/not-even-scientists-can-easily-explain-p-values/>
- ▶ <https://fivethirtyeight.com/features/science-isnt-broken/>
- ▶ <https://www.tandfonline.com/doi/pdf/10.1080/01973533.2015.1012991>

How about a frequentist answer these questions?

Before we start:

1. What's your best guess about θ ?
2. What's the probability that θ is greater than a half?

After we have observed some n trials and sample proportion $\hat{\theta} = Y/n$:

1. What's your best guess about θ ?
2. What's the probability that θ is greater than a half?

The Bayesian approach

- ▶ Bayesians also view θ as fixed and unknown.
- ▶ However, we express our uncertainty about θ using probability distributions.
- ▶ The distribution before observing the data is the **prior distribution**.
- ▶ Example: $\text{Prob}(\theta > 0.5) = 0.6$.
- ▶ Probability statements like this are more intuitive.
- ▶ This is subjective in that people may have different priors (we will also discuss objective Bayes).

The Bayesian approach

- ▶ Our uncertainty about θ is changed (hopefully reduced) after observing the data.
- ▶ The **Likelihood function** is the distribution of the observed data given the parameters.
- ▶ This is the same likelihood function used in a maximum likelihood analysis.
- ▶ Therefore, when the prior information is weak, Bayesian and maximum likelihood estimates are similar.
- ▶ Even in this case, the interpretations are different.

The Bayesian approach

- ▶ The uncertainty distribution of θ after observing the data is the **posterior distribution**.
- ▶ **Bayes theorem** provides the rule for updating the prior.

$$p(\theta|Y) = \frac{f(Y|\theta)\pi(\theta)}{m(Y)}.$$

- ▶ In words: Posterior $\propto$ Likelihood $\cdot$ prior.
- ▶ A key difference between Bayesian and frequentist statistics is that all inference is conditional on the single data set we observed Y .

Back to the example

- ▶ Say we observed $Y = 60$ successes in $n = 100$ trials.
- ▶ The parameter $\theta \in [0, 1]$ is the true probability of success.
- ▶ In most cases we would select a prior that puts probability on all values between 0 and 1.
- ▶ If we have no relevant prior information we might use the prior

$$\theta \sim \text{Uniform}(0, 1),$$

so that all values between 0 and 1 are equally likely.

- ▶ This is an example of an **non-informative prior**.

Posterior distribution

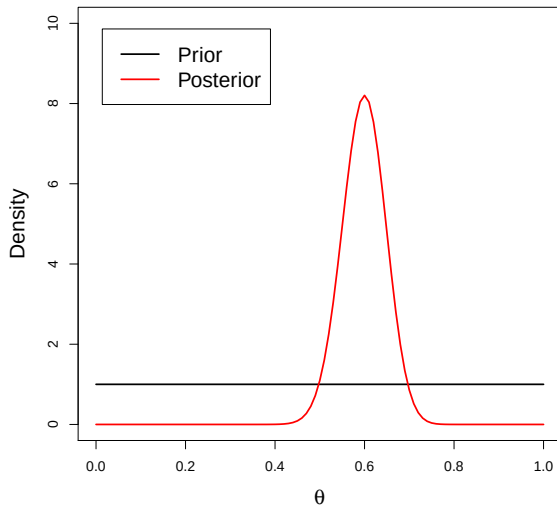
- ▶ The likelihood is $Y|\theta \sim \text{Binomial}(n, \theta)$.
- ▶ The uniform prior is $\theta \sim \text{Uniform}(0, 1)$.
- ▶ So, the posterior is

$$p(\theta|Y) \propto f(Y|\theta)\pi(\theta) \propto \theta^Y(1 - \theta)^{n-Y} \times \mathbb{1}(\theta \in [0, 1]).$$

- ▶ Then it turns out the posterior is

$$\theta|Y \sim \text{Beta}(Y + 1, n - Y + 1).$$

Bayesian learning: $Y = 60$ and $n = 100$



Beta prior

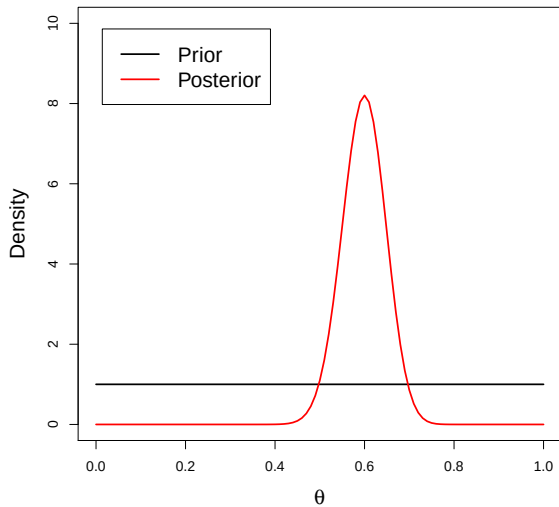
- ▶ The uniform prior represents prior ignorance.
- ▶ To encode prior information we need a more general prior.
- ▶ The beta distribution is a common prior for a parameter that is bounded between 0 and 1.
- ▶ If $\theta \sim \text{Beta}(a, b)$ then the posterior is

$$p(\theta|Y) \propto f(Y|\theta)\pi(\theta) \propto \theta^Y(1-\theta)^{n-Y} \times \theta^{a-1}(1-\theta)^{b-1} \mathbb{1}(\theta \in [0, 1]).$$

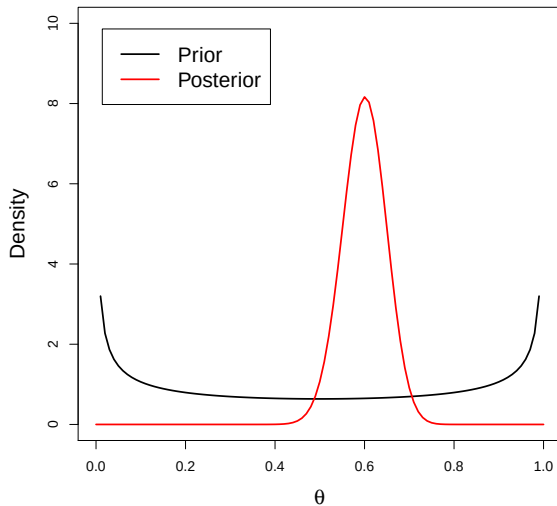
- ▶ Then it turns out the posterior is

$$\theta|Y \sim \text{Beta}(Y + a, n - Y + b).$$

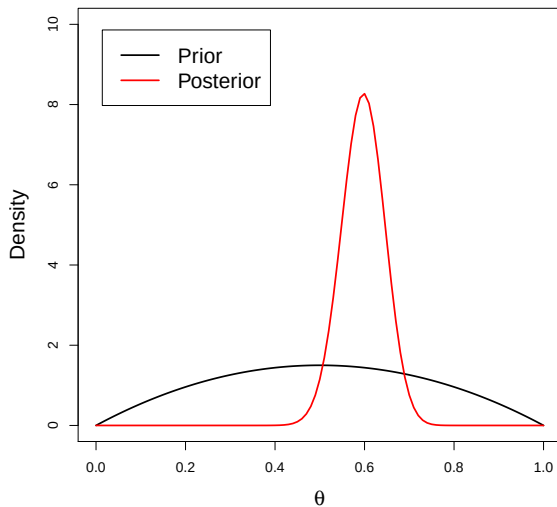
Prior 1: $\theta \sim \text{Beta}(1, 1)$



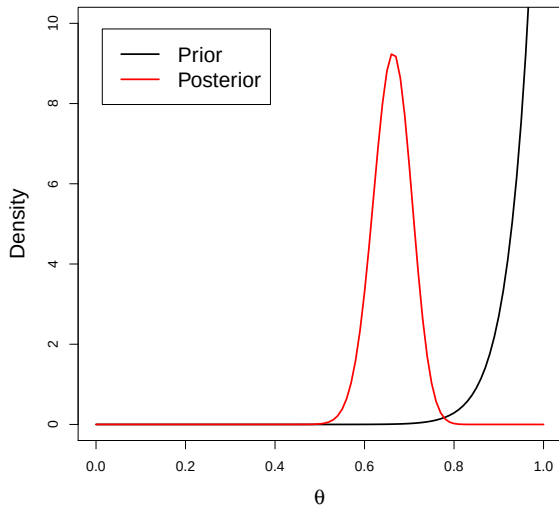
Prior 2: $\theta \sim \text{Beta}(0.5, 0.5)$



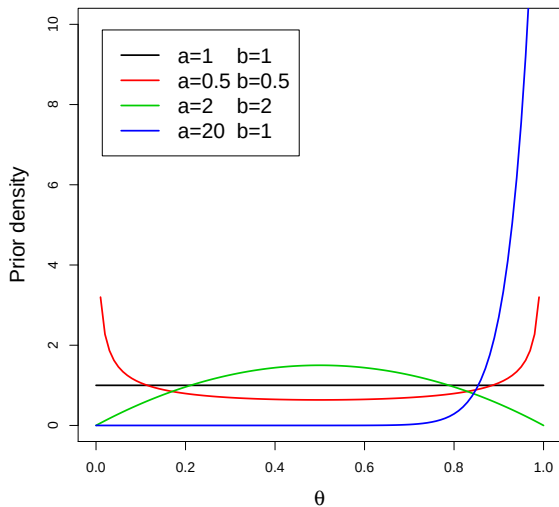
Prior 3: $\theta \sim \text{Beta}(2, 2)$



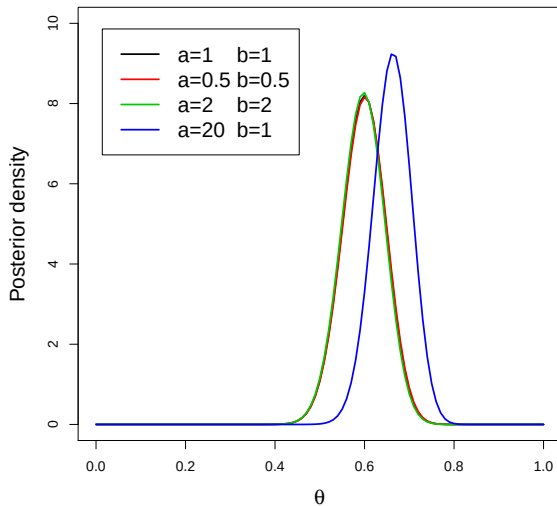
Prior 4: $\theta \sim \text{Beta}(20, 1)$



Plot of different beta priors



Plots of the corresponding posteriors



Sensitivity to the prior

a	b	Prior			Posterior		
		Mean	SD	$P > 0.5$	Mean	SD	$P > 0.5$
1	1	0.50	0.29	0.50	0.60	0.05	0.98
0.5	0.5	0.50	0.50	0.50	0.60	0.05	0.98
2	2	0.50	0.22	0.50	0.60	0.05	0.98
20	1	0.95	0.05	1.00	0.66	0.04	1.00

Summary

- ▶ The first three priors give essentially the same results.
- ▶ Say the objective is to test $\mathcal{H}_0 : \theta \leq 0.5$ versus $\mathcal{H}_A : \theta > 0.5$.
- ▶ In these three cases, we can say that, after observing the data, the probability of the null is only 0.02, and the alternative is 49 times more likely than the null.
- ▶ The final prior strongly favored large θ and gave different results.

Advantages of the Bayesian approach

- ▶ Bayesian concepts (posterior prob of the null) are arguably easier to interpret than frequentist ideas (p-value).
- ▶ We can incorporate scientific knowledge via the prior.
- ▶ Excellent at quantifying uncertainty in complex problems (e.g., missing data, correlation, etc.).
- ▶ In some cases (mainly for complex problems), the computing is easier.
- ▶ Provides a framework to incorporate data/information from multiple sources.

Disadvantages of Bayesian methods

- ▶ Less common/familiar.
- ▶ Picking a prior is subjective (we will study objective priors).
- ▶ Procedures with frequentist properties are desirable (we will study the frequentist properties of Bayesian methods).
- ▶ Computing can be slow or unstable for hard problems.
- ▶ Nonparametric methods are challenging.

Thank you!